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Pressure measurement during thermal runaway of lithium-ion batteries: Comparison of piezoelectric and piezoresistive pressure sensors

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ABSTRACT

Although lithium-ion batteries (LIBs) exhibit excellent energy storage characteristics, they harbor a potential risk due to their ability to undergo thermal runaway (TR). TR is accompanied by a rise in pressure and temperature, as well as possible secondary effects (e.g. explosions). A common mitigation strategy is to house LIBs in enclosures with a high pressure resistance (e.g. flameproof enclosures). The maximum pressure that occurs is therefore an important design criterion for such enclosures. Both piezoelectric (PE) and piezoresistive (PR) sensors can be used to measure the pressure. However, it is currently unclear whether one of the two sensor types is better suited for this specific measurement task. For comparative analysis, TR experiments using a LIB (3 Ah capacity, NMC-811 chemistry) are carried out in a flameproof enclosure using overheating as the failure mode. In addition to various temperature readings the pressure is recorded using both sensor types. Due to the high temperature load caused by the hot vent gases of the LIB during TR and the affinity of the PE sensors to so-called thermal shock, the measured pressure values are smaller than those measured by the PR sensors. By using PE sensors without a thermal protective layer (RTV) on the sensor membrane, the temperature can be identified as the decisive factor for the deviation. PR sensors should therefore be used for correct pressure measurement during the TR of a LIB. This study thus provides a direct performance comparison of PE and PR oil-filled sensors under thermal runaway and thermal shock conditions, recommending PR sensors as the more reliable choice for pressure assessment in lithium-ion battery safety applications. The study presents a methodologically sound evaluation framework by first validating the measurement system via controlled gas explosions, followed by testing TR under various gas atmospheres to enable a comprehensive comparison of sensor performance under realistic conditions. The vibration of the enclosure and the sensor design are also identified and discussed as disturbance variables of the pressure curve.

1. Introduction

Lithium-ion batteries (LIBs) have become an integral part of our everyday lives as a form of energy storage. They play a crucial role in the energy transition, with their deployment being essential for achieving the targeted reduction in greenhouse gas emissions by 2030. According to the European Green Deal, the objective is to cut emissions by 55 % compared to 1990 levels [1]. To this end, renewable energy sources are to greatly increase their share of total energy consumption compared to fossil fuels. The focus is on solar and wind as energy generators, as well as battery storage to compensate for the temporal shift between energy

generation and consumption [2]. In 2024, global cumulative installed battery capacity solely for electric vehicles reached 1 TWh for the first time, with LIBs being the most prevalent battery technology [3]. They are the predominant technology for stationary energy storage, as well as for use in electromobility and portable applications [4]. Projections indicate that stationary storage systems alone will reach a capacity of 1.6 TWh by the year 2030 [2]. Despite their excellent properties as energy storage systems, LIBs pose the risk of "thermal runaway" [5]. The sheer quantity of LIBs worldwide, even with estimated failure rates as low as 1:10 to 1:40 million (including safety equipment), indicates that accidents related to TR are not uncommon [37]. The term TR is used here to describe the phenomenon of significant self-heating, a

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Abbreviations

C ₃ H ₈	Propane
H ₂	Hydrogen
LIB	Lithium-ion battery
N ₂	Nitrogen
PE	piezoelectric
PR	piezoresistive
PRT	Pressure rise time
RTV	Room-temperature-vulcanizing
TR	Thermal runaway

consequence of exothermic reactions between different battery components. In addition to the emission of hot and toxic gases, sparks, and particle ejection, TR is characterized by a considerable increase in temperature and, in a confined systems, an increase in pressure. Secondary effects, such as explosion and fire, are also possible [7–9]. This is particularly dangerous if the TR occurs in potentially explosive atmospheres, as it can result in the ignition of the explosive environment [10, 11].

In order to mitigate the risk in such areas, types of protection such as flameproof enclosures can be used in accordance with the standard IEC 60079-0 [12]. According to the standard IEC 60079-1 a flameproof enclosure contains potential ignition sources, such as the LIB, in a way that prevents the transmission of flames, sparks, or excessive temperatures to the environment in the event of a fault. Since an explosion inside the enclosure and the associated increase in pressure is permitted on condition that the type of protection remains intact (permanent deformation or damage prohibited), the maximum pressure is a decisive design criterion [13,14]. In order to meet this criterion of pressure resistance, a sound knowledge base of the expected pressure values is required. Measuring the state variable pressure is not a novel topic, even within the field of explosion protection. Various pressure sensors have been tested with regard to their suitability for measuring dynamic, quasi-static and static pressures [7,14–16]. But the scope of the investigations extends beyond the mere use of sensors. For instance, several studies by Krause et al. are dedicated to detailed parasitic effects, such as those triggered by temperature or vibrations [15,16]. Furthermore, pressure measurements have also been successfully carried out during the TR of a LIB in various studies [7–9,17]. However, the large number of battery types and the heterogeneity of the test set-ups lead to a wide range of pressure values. In addition, the TR also represents a high thermal load, the influence of which on the measurement accuracy of pressure values remains to be investigated.

While pressure measurement during lithium-ion battery thermal runaway has been reported in various studies [7–9,17], these investigations primarily focus on characterizing the TR process itself rather than evaluating the suitability and reliability of different pressure sensor technologies under the extreme conditions encountered during TR events. The necessary properties that a pressure sensor should fulfill for this measurement task are not yet clearly defined, representing a significant knowledge gap, as the choice of sensor technology can critically affect measurement accuracy and reliability in safety-critical applications. To address this issue, this work presents the first systematic comparison between two widely used sensor types: piezoelectric (PE) and piezoresistive (PR) sensors. The present study addresses this gap by providing a comprehensive evaluation framework that systematically compares PE and PR sensor performance under authentic TR conditions. The methodology includes validation through gas explosions to ensure measurement system integrity, followed by TR experiments under various atmospheric conditions (air, nitrogen, and flammable gas atmospheres) to enable comprehensive sensor comparison under realistic operational stresses. The primary focus is on the dependence of the

pressure values on the positioning of the sensors and the ability of the two sensor types to deal with the thermal and mechanical load caused by the TR. In addition to the general comparison between PE and PR pressure sensors, two sensor models of the same measuring principle are compared with each other. The overall objective is to derive a substantiated recommendation for selecting the sensor type that best fits the intended application.

2. Materials and methods

The experiments presented here serve to compare PR and PE pressure sensors with regard to their suitability for pressure measurement during the TR of a LIB. For this purpose, a flameproof enclosure made of stainless steel and an aluminum-silicon alloy (type: 8264, company: R. STAHL AG) with a volume of approx. 38 l is used. The schematic view of the experimental setup can be seen in Fig. 2.1.

Within the enclosure, the LIB is positioned in a semi-circular cut-out on a copper block with a heating element inside (type: MFH14, company: Paul Rauschert Steinbach GmbH) and heated with constant power up to the TR. Two oscilloscopes (type: DL850, company: Yokogawa) and a data acquisition system (type: 34970A Data Acquisition/Data Logger Switch Unit, company: Keysight Technologies) are used to record the pressure, the cell voltage and current, various temperatures and the material strain. The strain values are used to verify the pressure values and are not discussed in detail [7,14]. The temperatures are recorded using thermocouples of the type K1/0.711 from RS Components GmbH and are labeled individually (thermometers). If a gas atmosphere other than air is to be created inside the enclosure, this can be done before the start of the test using mass flow meters for gases (type: F-112AC; company: Bronkhorst High-Tech BV). The mixture composition is checked using an oxygen analyzer of the type 570A Portable Oxygen Analyser (company: Servomex GmbH). Sufficient purging also ensures a tenfold gas exchange in the enclosure.

2.1. Test configurations

As part of the investigations presented here, two pressure sensor types are tested under various loads to determine the key data for their suitability for pressure measurement of the TR of a LIB. Four different gas atmospheres are used inside the enclosure for this purpose. These are designated below with the first letter of the gas (A, H, P, and N) and the suffix "TR" for tests with LIB and "gas" for pure gas explosions. The utilization of an air atmosphere (tests A-TR) denotes the "base case," which is the most common in reality. The catalytic effect of atmospheric oxygen maximizes the severity of the TR and enables secondary effects such as a fire and explosion of the reaction gases [7]. In order to suppress the influence of atmospheric oxygen, the enclosure is flooded with nitrogen (N₂, N-TR). This represents the least critical case in terms of the resulting pressure and temperature load. Conversely, combustible gas-air mixtures are used to increase the load on the pressure sensors even further compared to air and to imitate a real explosive atmosphere. This case is relevant for the use of LIB in potentially explosive areas. Propane (C₃H₈, P-TR) is used as a representative of gas group IIA and hydrogen (H₂, H-TR) of gas group IIC (IEC 60079-0) in their stoichiometric mixture compositions of 4.6 ± 0.3 vol.-% and 31 ± 1 vol.-%, respectively [13]. While a H₂-air explosion causes a particularly rapid increase in pressure, C₃H₈-air explosions lead to a high maximum pressure [7,29]. The employment of these gas atmospheres covers a range of loads, in which the vast majority of possible TR scenarios occur.

2.2. Pressure sensors

The following section provides a brief overview of the pressure sensors used and their respective properties, as well as the positioning of the sensors.

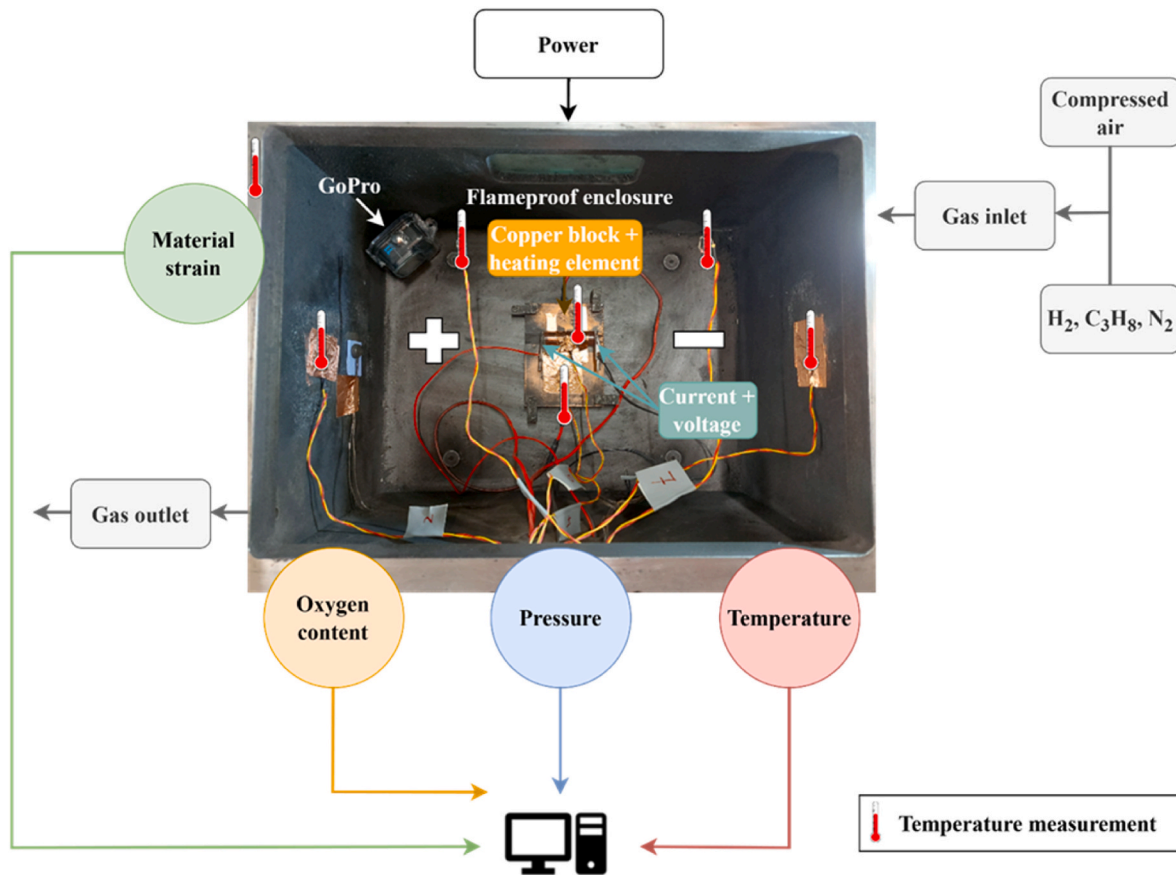


Fig. 2.1. Depiction of the experimental setup, with plus and minus indicating the positive/negative terminal side of the LIB and thermometers indicating locations for temperature measurements (extended from Ref. [7]).

2.2.1. Used sensors and their properties

A total of five PR and six PE sensors are used, all of which are part of the Kistler Group product range. Two different PR oil-filled sensors (4080B and 4011A with M14 adapter 7539A) and two different PE sensors (601CAA and 6031, both with M10 adapter 6503C0A) are utilized. All sensors are mounted flush with the underside of the enclosure lid. The PR sensors 4080B and 4011A differ in their design. The 4011A type allows the sensor membrane to be mounted flush with the underside of the lid, while the membrane of the 4080B type is slightly recessed, creating a cylindrical cavity that can lead to acoustic oscillations (see Section 3.3.3). In addition to a pressure output, both sensors have a temperature output (indirect measurement from the Wheatstone bridge), allowing for the temperature of the sensor electronics to be monitored. This is a critical aspect for evaluating the measurement uncertainty, as it depends on compliance with the calibrated temperature measuring range. The PE sensor types 601CAA and 6031 were selected on the basis of their opposing characteristic profiles. Type 601CAA sensors exhibit reduced thermal shock sensitivity and heightened acceleration sensitivity, while type 6031 sensors show the reverse characteristics. All PE sensors used here are usually equipped with a thermal protective layer made out of silicone (room-temperature-vulcanizing, RTV, 1 mm thick) on their sensor membrane (see Section 3.3.1) [16]. For the coating, a RTV silicone rubber of type RTV102 (manufacturer: Momenite) was applied. The coating procedure utilized the dedicated Kistler RTV Tool (Type 1300A195) to ensure a uniform layer thickness and consistent coverage. This application follows the recommended method described in the 601C/603C sensor instruction manual (see pages 40–41) [21]. The individual sensitivities for each sensor in the range of $-37 \text{ pC} \cdot \text{bar}^{-1}$ (601CAA) resp. $-14 \text{ pC} \cdot \text{bar}^{-1}$ (6031) are adjusted according to the measuring range of 0–20 bar [19,20]. Due to

the thickness of the silicon chip installed in the sensor, which is decisive for its pressure measurement range, it is not possible to adjust the measuring range for PR sensors. Both sensors with a measuring range of 0–10 bar (4080B and 4011A) and those with a larger measuring range of 0–20 bar (4080B) are used. Table 2.1 offers a comprehensive overview of the relevant properties of the pressure sensors. To differentiate between the two PR and PE sensor models, the number "1" (type 601CAA or 4080B) or "2" (6031 or 4011A) is appended after the sensor type (PE or PR) and the position on the lid ($x = 1 \dots 6$).

The "LabAmp" charge amplifier (type 5167Ax0; manufacturer: Kistler Group) with a set measuring range of 20 bar and a long time constant ($>10,000 \text{ s}$) is utilized for all PE sensors [23]. The time constant referred to as "DC Long" is selected to ensure the detection of low-frequency components of the measurement signal [24]. While the PR pressure sensor 4080B does not require an additional amplifier, the amplifier 4624AK (manufacturer: Kistler Group) is used to amplify the voltage signal of the sensor 4011A.

Table 2.1
Characteristics of the pressure sensors [20–22,36].

Property	piezoelectric (PE)		piezoresistive (PR)	
Sensor type	601CAA	6031	4080B	4011A
Name	PEx.1	PEx.2	PRx.1	PRx.2
Measuring range/bar	0 ... 250	0 ... 250	0 ... 10/20	0 ... 10
Sensitivity	$-37 \text{ pC} \cdot \text{bar}^{-1}$	$-14 \text{ pC} \cdot \text{bar}^{-1}$	$420/210 \text{ mV} \cdot \text{bar}^{-1}$	$1000 \text{ mV} \cdot \text{bar}^{-1}$
Acc. sensitivity ax./mbar $\cdot \text{g}^{-1}$	2	<0.1	<0.4	<0.2

2.2.2. Positioning of the pressure sensors

Fig. 2.2 shows a depiction of the flameproof enclosure lid, illustrating the positioning of the pressure sensors relative to each other and to the explosion site or positive terminal of the LIB. The illustration is not intended to convey exact dimensions, but rather to clarify the usefulness of the various sensor positions for the analytical comparison of the sensor types. Sensors PE1.1- to PE5.1+ are model 601CAA sensors, while PE6.2+ corresponds to model 6031. In a similar manner, sensors PR1.1+ to 4.1- are model 4080B sensors, whereas model 4011A was selected for PR5.2-. As can be seen in Fig. 2.2, the pressure sensors on the lid are divided into "+" and "-". This differentiation is important because the behavior of a round cell, as used in this study, differs depending on the terminal side. It is not yet clear whether this makes a significant difference for pressure measurement during TR (see Section 3.1). The placement of pressure sensors of both types on both terminal sides facilitates the analysis of the positional dependence of the pressure measurement data. For this purpose, two sensors with PR (PR2.1+ and PR3.1-) and PE measurement principles (PE2.1- and PE3.1+) have been placed directly opposite each other at the corners of the lid viewing glass. Additionally, a comparison of sensor types can be achieved by using adjacent sensors on the same terminal side. The positions directly at the positive and negative terminal side also permit a direct comparison of both sensor types in mirror-image positioning.

Further, a PE pressure sensor (601CAA) is installed in a blind hole. By positioning the blind hole in front of a sensor of identical design, the impact of parasitic signals on the measurement signal of the through hole can be evaluated [15]. For a selection of the tests presented here, pressure sensors without RTV are used at sensor positions PE2.1-, PE3.1+, and PE6.2+.

2.3. LIB key data

In all destructive tests involving a LIB, the INR18650HG2 round cell from LG Chem with a nominal capacity of 3 Ah and NMC-811 cell chemistry (specifically, $\text{Ni}_{0.84}\text{Mn}_{0.06}\text{Co}_{0.1}$) was used [25,26]. All cells were charged to a voltage of 4.2 V, corresponding to a state of charge of 100 %, before the start of the test. This LIB was selected based on the reproducibility of the pressure values in previous tests, as reported by Spörhase [27].

2.4. Verification of the test setup using gas explosions

To assess the suitability of the test setup for measuring explosion

pressures due to the TR of a LIB, verification procedures 4.3 and 4.4 according to EN 15967 are referred to as a guide. In accordance to Annex A ("Verification of values for maximum explosion pressure"), Table A.1, H_2 in its stoichiometric mixture composition with air (31 ± 1 vol.-%) is used as the fuel gas [28]. The mixture is ignited with a spark plug, which is placed at the same location where the positive terminal of the LIB will later be located, so that the ignition is comparable. A comparison of the mean value of all sensor signals (PE and PR) from three tests of 7.94 ± 0.13 bar (absolute pressure) with the target value according to Table A.1 of 7.90 ± 0.30 bar demonstrates the suitability of the test setup presented here for measuring explosion pressures [28]. Based on the maximum rise rate of $1200 \text{ bar} \cdot \text{s}^{-1}$, when compared to the literature data reported by Chemsafe of $1539 \text{ bar} \cdot \text{s}^{-1}$, as well as the pressure rise time (PRT) of approximately 7 ms, it can be further concluded that the dynamics of both sensor types meet the requirements for dynamic pressure measurement set out here [7,29]. Consequently, the setup can be employed for all processes that are slower than an H_2 -air explosion, such as the TR of a LIB.

3. Theory

The following sections provide an introduction to the necessary theoretical background on LIB thermal runaway, flameproof enclosure, and the piezoelectric and piezoresistive effects.

3.1. The thermal runaway

The term "thermal runaway" is used to describe the process of excessive heating of the LIB due to exothermic decomposition reactions occurring within the cell [7,8]. The decomposition of the cell components is accompanied by gas emissions, which occur repeatedly during the overheating process of the LIB. The decomposition of the metal oxides in the cathode releases oxygen, which catalyzes the oxidation of the electrolyte and thus the release of further gases [30]. It is important to note that the hot reaction gases are initially generated inside the cell and do not escape to the outside. This only occurs when the internal overpressure exceeds the maximum permissible pressure of the safety valve at the positive terminal of the LIB. The release of gas in a closed system is accompanied by an increase in temperature and pressure. In addition, hot particles are ejected from the interior of the cell on the positive terminal side. As some of the released gases are highly flammable, secondary effects such as fire or explosion can occur in an oxygen-rich atmosphere. These further increase the pressure and

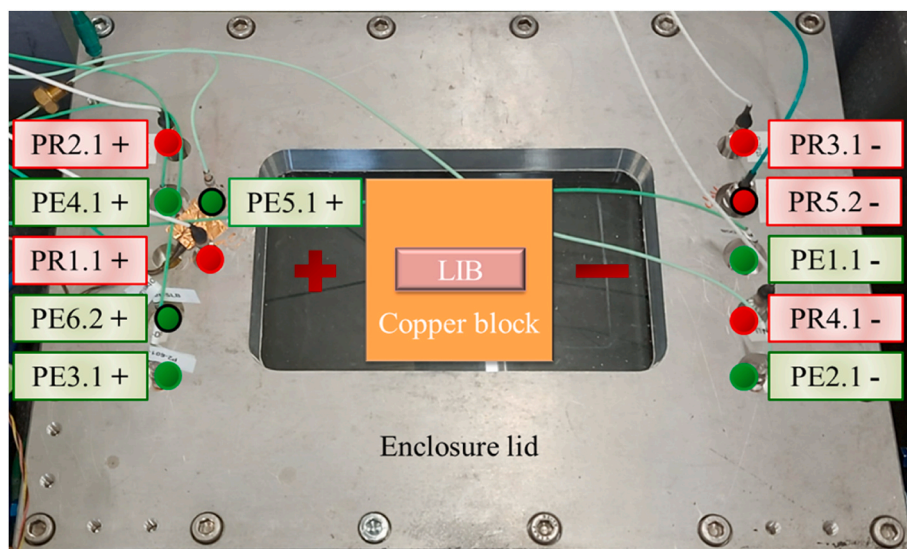


Fig. 2.2. Positioning of the PR or PE pressure sensors on the enclosure lid. The relative position to the terminals of the LIB is marked with a "+" or "-".

temperature, with the thermal load on the positive terminal side being more intense than on the negative terminal side. The high inhomogeneity of gas emission from the LIB can result in significant fluctuations in local pressure values [7]. As the maximum values that occur further depend on both the LIB and the surrounding system (enclosure, gas atmosphere), this results in a wide range of possible pressure and temperature values (i.e., <3 bar to >200 bar; >900 °C) [7, 8,10,17]. In general, TR is considered to represent a quasi-static process, with PRT ranging from <20 ms to >100 ms [7]. To assess the material stress caused by a TR in a pressure-resistant enclosure, the standard IEC 60079-1 specifies that the peak pressure as well as the PRT are to be used. The PRT denotes the period during which the pressure increases from 10 % to 90 % of its maximum value. Thermal load is not taken into account by the standard [13].

3.2. The type of protection flameproof enclosure

The flameproof enclosure is classified, according to IEC 60079-0, as one of the types of protection used in hazardous areas [12]. It is an enclosure designed to prevent the transmission of an internal explosion to the surrounding explosive atmosphere by means of appropriately dimensioned gaps, sufficient wall thickness, and limitation of the external surface temperature. The high pressure resistance provided by the wall thickness is one of the key design criteria. Compliance with these requirements is verified through test procedures in accordance with IEC 60079-1, which are based on the maximum pressure measured and the PRT during specified gas explosion tests [13,14].

3.3. Characteristics of the pressure sensors

The measurement of pressure inside the enclosure can be performed using several sensor types, with the choice of sensor depending on the expected load on the sensor. PR pressure sensors - depending on the precise measurement technology (e.g., oil-filled) - can be used to measure both static (time-invariant, e.g., water pressure) and dynamic pressure (time-variant, e.g., gas explosions) [7,19,31]. Typically, such sensors consist of semiconductor resistors arranged in a Wheatstone bridge. When subjected to pressure, a mechanical stress is induced, which in turn leads to a change in electrical resistance. This phenomenon is known as the piezoresistive effect. The change in resistance alters the bridge voltage. This voltage signal, on the order of millivolts (mV), is proportional to the pressure. Sensors that exploit the piezoresistive effect are characterized by high sensitivity, allowing for the detection of small pressure changes. Additionally, they are robust and compact [7]. A disadvantage is that the measurement range is fixed due to the sensor design and cannot be flexibly adjusted to the required measurement range [15,16]. Due to their low bandwidth and low natural frequencies, PR sensors are often incapable of measuring highly dynamic, i.e., rapidly changing, pressure [32,33].

Generally, PE pressure sensors can be employed to capture such pressure variations [21]. The piezoelectric effect is based on the generation of an electric charge resulting from the asymmetry of charge centers. The unit cell of the crystals used is structured in such a way that deformation caused by an applied load (e.g., pressure) leads to a displacement of the charge centers (asymmetry). Since these no longer coincide and thus do not compensate each other, a charge proportional to the pressure can be extracted from the crystal [7,15]. This charge is very small (picocoulomb, pC), which complicates the handling of the sensors and associated measurement technology such as cables and charge amplifiers [15,32]. However, the use of sensors based on the piezoelectric effect enables the measurement of highly dynamic processes due to their short response time and flexibly adjustable measurement range [7]. As the dynamics of the TR are classified as quasi-static, it remains unclear which sensor type is better suited for the measurement task investigated here.

3.3.1. Influence of transient temperatures

Not only pressure itself but also temperature is a critical state variable for pressure measurement. During the TR, the temperature changes dynamically and can reach high values exceeding 900 °C [8,17]. Transient temperatures have little effect on the measurement accuracy of PR oil-filled sensors due to their low temperature sensitivity. However, for PE sensors dynamic temperature gradients present a significant measurement challenge [32,33]. The response of such a sensor to a dynamic change in sensor and diaphragm temperature is referred to as thermal shock. Thermal shock causes a temperature-induced alteration of the measurement signal, making accurate pressure detection impossible. For example, a rapid temperature increase can lead to an apparent pressure decrease due to deformation of the sensor diaphragm and the resulting discharge of the crystal [21]. Due to the differing thermal expansion coefficients of the materials composing the PE sensor, the elastic force between the sensing element and the sensor housing changes under dynamic temperature gradients [32,34]. A characteristic feature of thermal shock is that the transient measurement signal, after reaching its maximum, rapidly drops into the negative pressure range and only slowly returns to its initial value [16,21]. RTV applied to the sensor diaphragm can delay thermal shock effects, thereby improving pressure measurement accuracy during transient thermal events. According to Krause et al., an ideal coating thickness is approximately 1 mm [16]. However, this concept has not yet been applied to the slower TR, which involves multiple temperature-generating stages.

3.3.2. Acceleration sensitivity

Due to the large mass and resulting inertia of typical enclosures of the explosion protection type "flameproof enclosure", a pressure wave inside the enclosure induces vibrations upon impact with the wall. The amplitude of these vibrations depends on the material and geometric properties of the enclosure. Since the pressure sensors are mounted on the enclosure, all vibrations of the enclosure are transmitted to the sensors themselves. The inertial forces acting on the sensor result in its acceleration, which can consequently generate a superimposed interference signal. This effect is referred to as acceleration sensitivity and its magnitude depends on the sensor design [15]. Since piezoelectric elements often have a relatively large mass and size, PE sensors are characterized by a pronounced acceleration sensitivity [32]. In contrast, PR sensors are less sensitive to acceleration (see Table 2.1). To assess the impact of acceleration-induced signals, PE sensors can be mounted in blind holes. This isolates them from the enclosure interior, preventing detection of the pressure rise. However, enclosure vibrations still affect the measurement signal. The vibration frequency components can be identified via Fast Fourier Transform (FFT). By applying suitable filters to the signals from standard through-hole sensors, this interference can be compensated, restoring the useful signal [15].

3.3.3. Acoustic resonance

Ideally, pressure sensors should be flush-mounted (see Fig. 3.1, a)). If this cannot be ensured, a dead volume forms in front of the sensor membrane, consisting of a usually cylindrical through-hole and a cavity (see Fig. 3.1, b), white). When a rapid pressure increase occurs at the beginning of the through-hole, causing fast pressure fluctuations, the gas in the cavity begins to compress and expand. This leads to the formation of standing waves, i.e., acoustic oscillations. These superimpose on the useful signal as parasitic interference and can, in the worst case, cause signal distortions such as pressure overshoots, with the result that quantities like the maximum pressure may not be determined accurately. The excited natural frequency of the acoustic oscillations mainly depends on the geometric parameters of the through-hole as well as the properties of the gas, such as its molar mass, which is excited in the cavity [35].

As an example, Fig. 3.1 c) shows the result of the simulation of the natural frequency of a stoichiometric H₂-air mixture at room temperature (25 °C) within the dead volume depicted in Fig. 3.1 b). For the

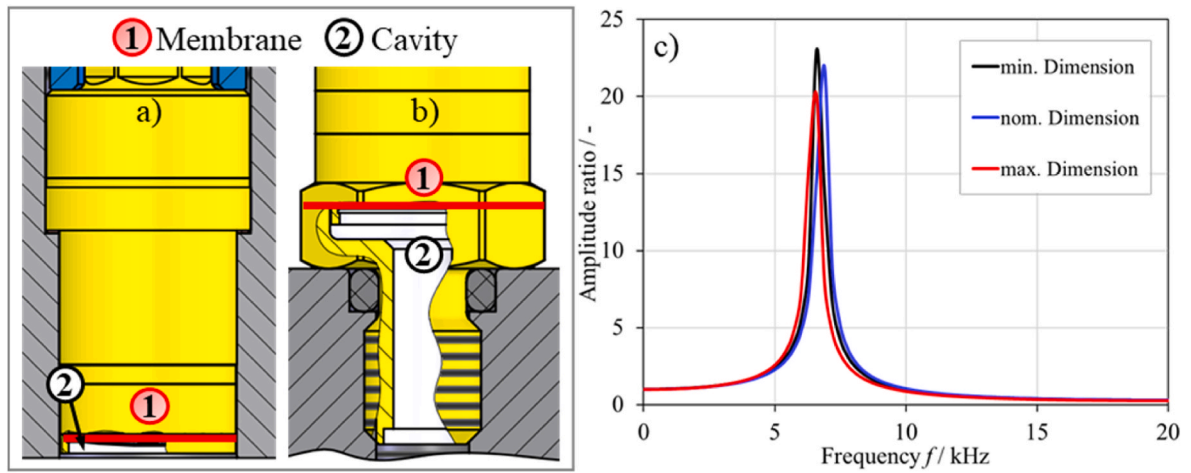


Fig. 3.1. Schematic representation of membrane position (red) and dead volume (white) of a) type 4011A (flush mounted), b) type 4080B and c) the corresponding acoustic resonance frequency of a stoichiometric H_2 -air mixture at 25 °C (type 4080B).

nominal dimensions of the dead volume, a natural frequency of 7 kHz can be obtained (see Fig. 3.1, c), blue). Minor uncertainties due to the cavity's manufacturing tolerance are denoted by the "min./max. dimension" in blue and red and have minimal influence on the natural frequency. When the air content is increased, the natural frequency decreases to approximately 3.5 kHz (100 vol.-% air). Given the comparable molar masses of air, N_2 , and a stoichiometric C_3H_8 -air mixture, the acoustic oscillations also lie within the same frequency range.

4. Results and discussion

Fig. 4.1 a) illustrates an idealized pressure curve as a function of time. For the targeted comparison of the pressure sensor types, the parameters PRT and peak pressure p_{ex} are considered. Both are indicated in

Fig. 4.1 a). In contrast, Fig. 4.1 b) presents the actual measurement signals from a representative experiment of a TR, which is superimposed by a gas explosion (C_3H_8 -air). The additional gas explosion does not affect the qualitative characteristics of the pressure profile; consequently, this test configuration can be used as an example. Fig. 4.1 b) thus serves merely to illustrate the main issues:

1. Drift between PR and PE sensor signals
2. Collapse of the measurement signals from PE2.1- (green) and PE3.1+ (red)
3. Oscillations in all pressure curves, here represented by the sensor installed in the blind hole (PE5.1+, black)

The following sections analyze and discuss in detail the occurring

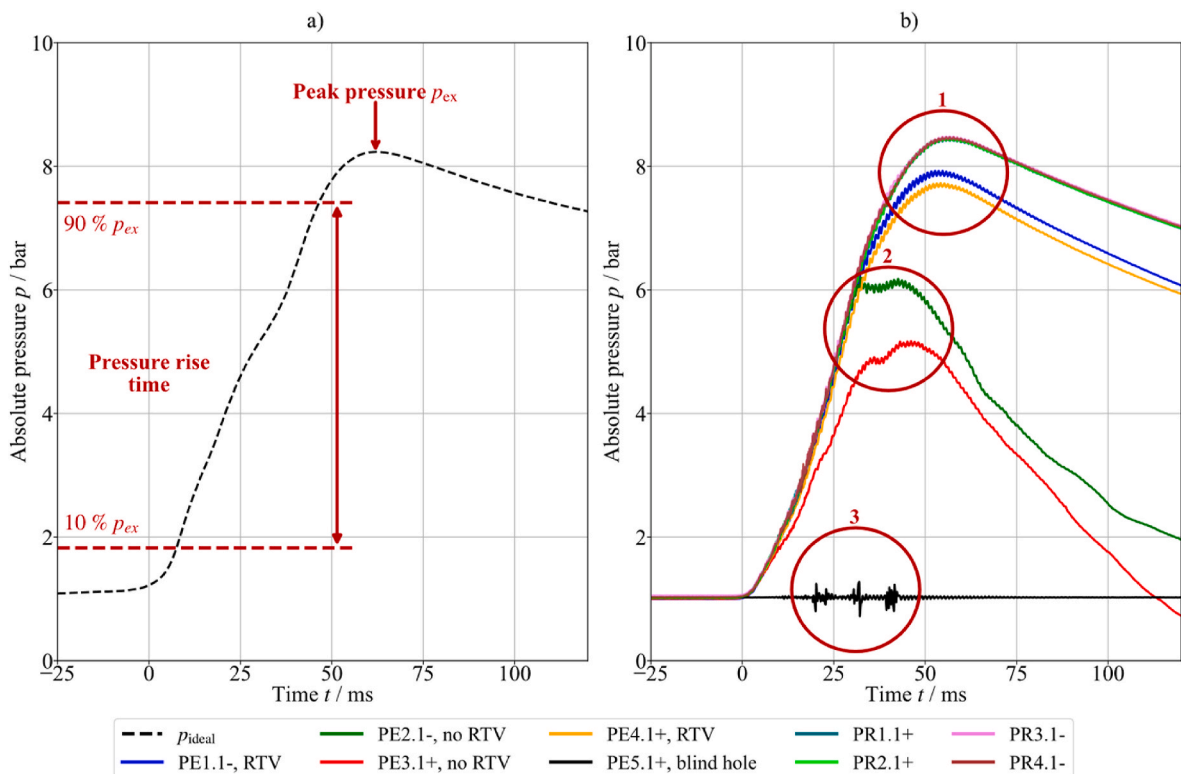


Fig. 4.1. Comparison of the idealized a) and actual time-resolved pressure curve; b) of a C_3H_8 -air explosion and TR (P-TR).

effects and disturbances depending on the sensor type, the underlying causes, and possible solutions. All experimental data were processed using a second-order Butterworth low-pass filter with a cutoff frequency of 5 ± 0.5 kHz, in accordance with IEC 60079-1 [13]. Table 4.1 offers an overview of all experiments conducted. For each test configuration, three experiments were conducted unless stated otherwise, and the mean pressure values of the PE and PR sensors are discussed. Additionally, values for the PRT, as well as the material strain of the enclosure sidewall, ϵ_{ex} , are provided. Values for PE2.1- and PE3.1+ are not reported in Table 4.1, as tests with and without RTV were conducted for these sensors. Sensors without RTV are excluded from the mean of the PE pressure sensors since they would distort the results. In the case of the sensor that has been installed in the blind hole, PE5.1+, the maximum values are presented instead of the mean values to provide an impression of the intensity of the interference. Given that the load under air and N₂ atmosphere is lowest and thus encompassed by the maximum values of C₃H₈- and H₂-air, recording of the PE5.1+ sensor signal for A-TR and N-TR was omitted. Furthermore, the sensors PE6.2+ (model 6031) and PR5.2- (model 4011A) are exclusively employed for comparison of specific qualitative properties of the individual models, which is why quantitative measurement values are not provided in Table 4.1.

Additionally, the PRT is calculated as the mean of all PR sensor signals. In the case of N-TR, the mean value only consists of two experimental values. For ϵ_{ex} , the mean value from two to three trials (marked with a) is reported, as the measurand was not available for all experiments. Since the strain values serve solely as a parameter to verify the pressure data and only need to be within a comparable range to literature values, reporting the mean from two trials is considered adequate. The same logic applies to the PRT values.

4.1. Position dependency of the pressure values

Initially, based on the results from Table 4.1 as well as exemplarily on the pressure curves from Fig. 4.1 b), it can be observed that, regardless of the gas atmosphere, the measured peak pressure values of the PE sensors are consistently lower than those of the PR sensors. Based on findings of prior research conducted at the Physikalisch-Technische Bundesanstalt (PTB), where a PR pressure sensor (type: 4045A100; company: Kistler Group) was utilized within the same flameproof enclosure, the following values for peak pressure and material strain are

Table 4.1
Overview of the mean values (MV) for p_{ex} , PRT, and ϵ_{ex} in air, N₂, C₃H₈-air, and H₂-air

Measurand	Sensor	A-TR	N-TR	P-TR	H-TR
$p_{ex,PE}/\text{bar}$	PE1.1-	4.67 ± 0.09	2.36 ± 0.10	7.91 ± 0.06	7.55 ± 0.11
	PE4.1+	4.10 ± 0.49	2.28 ± 0.07	7.58 ± 0.17	7.16 ± 0.24
	MV	4.45 ± 0.22	2.32 ± 0.08	7.67 ± 0.14	7.22 ± 0.22
	PE5.1+	/	/	0.26 ± 0.06	0.11 ± 0.06
$p_{ex,PR}/\text{bar}$	PR1.1+	5.56 ± 0.05	2.53 ± 0.10	8.52 ± 0.06	7.90 ± 0.06
	PR2.1+	5.56 ± 0.04	2.52 ± 0.10	8.54 ± 0.07	7.91 ± 0.05
	PR3.1-	5.53 ± 0.05	2.53 ± 0.10	8.52 ± 0.09	7.93 ± 0.06
	PR4.1-	5.50 ± 0.04	2.48 ± 0.11	8.52 ± 0.09	7.90 ± 0.07
	MV	5.54 ± 0.04	2.51 ± 0.10	8.52 ± 0.07	7.91 ± 0.05
	PRT/ms	102.5 ± 7.6	119.2 ± 6.8 ^a	30.7 ± 3.1	7.1 ± 0.1
$\epsilon_{ex}/\mu\text{m} \cdot \text{m}^{-1}$	MV	288 ± 11	92 ± 5 ^a	427 ± 1 ^a	395 ± 5 ^a

^a Mean value from two instead of three experiments.

considered as reference [7]:

- A-TR: $p_{ex} = 5.83 \pm 0.88$ bar, $\epsilon_{ex} = 291 \pm 76 \mu\text{m} \cdot \text{m}^{-1}$
- P-TR: $p_{ex} = 8.21 \pm 0.04$ bar, $\epsilon_{ex} = 423 \pm 5 \mu\text{m} \cdot \text{m}^{-1}$
- H-TR: $p_{ex} = 7.98 \pm 0.09$ bar, $\epsilon_{ex} = 406 \pm 12 \mu\text{m} \cdot \text{m}^{-1}$

For N-TR, a value of $p_{ex} = 2.87 \pm 10$ bar ($\epsilon_{ex} = 115 \pm 12 \mu\text{m} \cdot \text{m}^{-1}$) obtained from a single experiment at the PTB is used as a reference point. A comparison of the mean values in Table 4.1 with the reference data shows good agreement for the PR sensors, while the PE sensor values are consistently lower. Additionally, the measured strain values align with literature data, further validating the PR pressure measurements. These findings suggest that the PR sensors provide a reliable representation of the actual pressure. Since the four PR sensors show no significant deviations in peak pressures, the results suggest homogeneous pressure wave propagation during the TR. Thus, it can be concluded that the maximum pressure values do not vary spatially. Consequently, the observed deviation of the PE pressure values towards lower values cannot be attributed to the applied pressure but rather suggests the presence of a parasitic influence.

4.2. Effect of temperature-generating processes

Since drift at the positive terminal side is always more pronounced than at the negative terminal side (see Fig. 4.1, b)), it is suspected that this drift is thermally induced. To verify this hypothesis, the protective RTV was removed from the PE2.1- and PE3.1+ sensors. Fig. 4.2 shows the measurement results of the PE pressure sensors (PE1.1- to PE4.1+) from one representative experiment for each gas atmosphere. Panels a) through d) display A-TR, N-TR, P-TR, and H-TR, respectively. In all test configurations, a pronounced thermal shock occurred at sensors PE2.1- and PE3.1+ (without RTV), which led to a reduction in pressure values down to values below 0 bar. This effect is particularly pronounced for PE3.1+ ($PE3.1+ < PE2.1-$) since thermal load at the positive terminal side of the battery occurs earlier and more intensely (see Section 3.1). For the positive terminal side pressure sensor without RTV (PE3.1+), a collapse in the measurement values is observed early in the pressure rise. Compared to PE1.1- (blue), which experiences the lowest thermal load, PE3.1+ (red) shows only half the pressure when exposed to an air atmosphere (see Fig. 4.2, a)), whereas under H₂-air conditions (see Fig. 4.2, d)), the pressure reduction is, on average, only about 14 %.

A key difference between the TR of a LIB and a gas explosion lies in the progression of the process. As described in Section 3.1, TR initially begins with a rapid development and expansion of hot reaction gases before secondary effects such as fire or explosion occur. The gas release is inhomogeneous, which is why pressure values and local heat input can vary significantly (see Section 3.1). This is particularly visible in the measurement results of both the air and N₂ atmospheres (Fig. 4.2a) and b)). In the case of PE3.1+ (red), a slightly undulating shape is evident, which can be explained by the inhomogeneous temperature distribution in the gas and is also reflected in the measurement data from the thermocouples in the free gas volume of the enclosure. The slower the overall process, the more the gas evolution of the battery becomes dominant, leading to greater fluctuations in heat input. Since the thermal shock is caused by the different thermal expansion coefficients of the pressure sensor materials, prolonged exposure to a thermal load results in a greater change in the elastic force between the sensor components and ultimately a stronger influence on the measurement signal (see Section 3.3.1). Additionally, the amount of oxygen inside the enclosure is also decisive for the total quantity of gas released (see Section 3.1). Accordingly, among all experimental configurations presented here, the largest amount of reaction gas is produced under air atmosphere conditions.

The use of the model 6031 without RTV (as PE6.2+), which exhibits higher thermal shock sensitivity compared to the model 601CAA, further supports the assumption of temperature-induced pressure

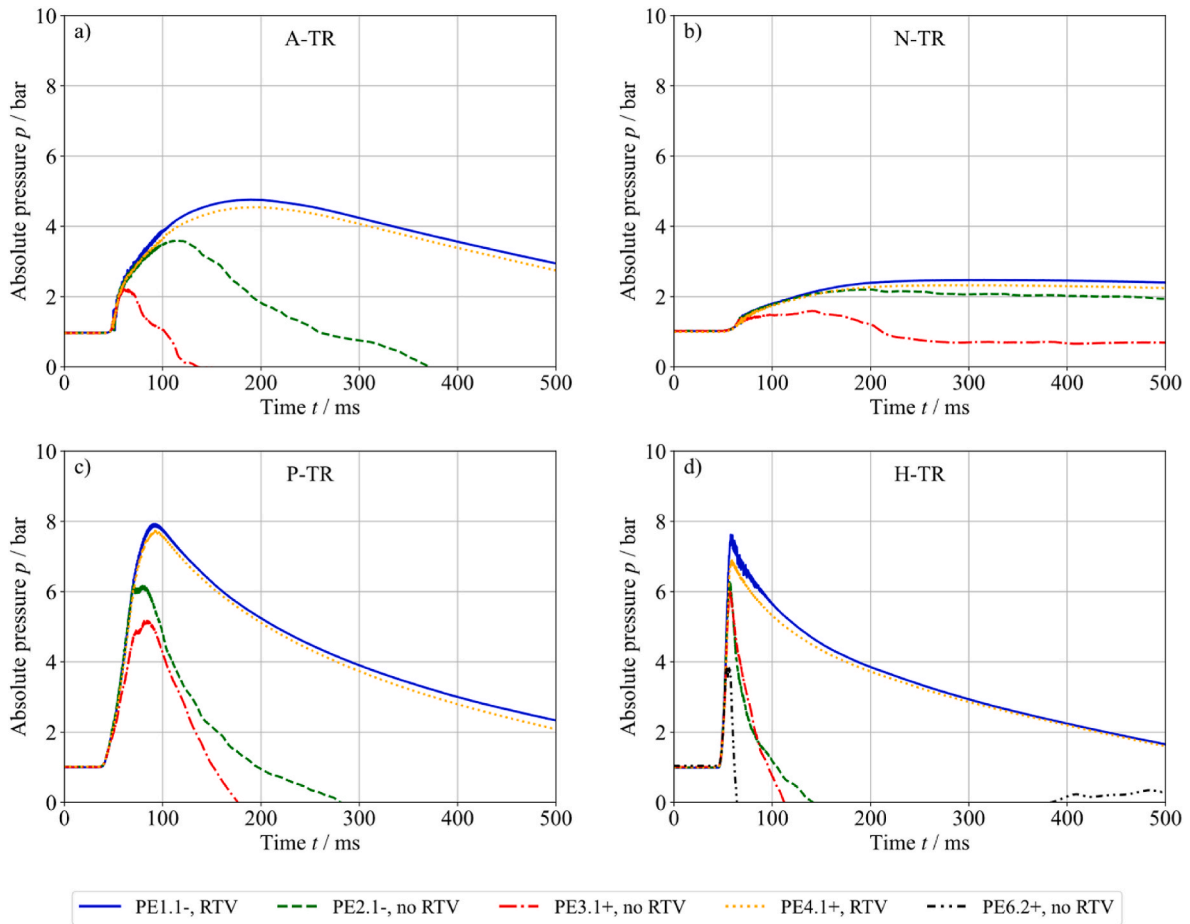


Fig. 4.2. Comparison of PE sensors with and without RTV for a) A TR, b) N TR, c) P TR, and d) H TR.

reductions. This is evident in Fig. 4.2, d) for H₂-TR. The thermal shock is so dominant that the pressure barely increases and instead almost immediately drops steeply into the negative range. Even when using RTV, a difference in PE measurement values between the positive and negative terminal side can generally be observed, such that PE4.1+ < PE1.1- in all cases. The deviation between PE1.1- and PE4.1+ manifests within the first milliseconds of the pressure rise (see Fig. 4.3) and is therefore attributable to gas venting. The delay of the thermal shock due to the RTV is thus insufficient to compensate for the significantly

prolonged thermal load from TR compared to a pure gas explosion, preventing accurate detection of the pressure profile. Even in the extreme case of high dynamics (H₂-TR, Fig. 4.2, d)), a slight drift can be observed before the maximum pressure is reached. Despite the suitability of PE pressure sensors for measuring explosion pressure in fuel gas-air explosions, they prove to be a less favorable choice compared to PR sensors in this scenario.

4.3. Influence of acceleration sensitivity

Fig. 4.4 exemplifies another interference phenomenon in the form of oscillating measurement signals. Observed consistently across all experiments, these oscillations stem from vibrations of the flameproof enclosure itself.

Due to its high mass and associated inertia, pronounced natural vibrations develop and are transmitted to the bolted sensors. As a consequence, parasitic signals are generated, which superimpose the true pressure signal. To isolate these undesired components, the measurement signal from the blind-hole-mounted sensor PE5.1+ was subjected to FFT-based frequency analysis (see Section 3.3.2). Fig. 4.4 a) presents the resulting spectrum during an H₂-TR experiment. Owing to the rapidity of this explosion, a wide frequency spectrum is excited, thereby revealing the enclosure's vibrational response characteristics. As can be seen, the spectrum is dominated by a 1.1 kHz frequency corresponding to the fundamental mode. Associated higher-order modes can also be identified at approximately 2.2, 3.2, and 4.2 kHz as well as a concise frequency peak at 1.9 kHz. This enclosure resonance is evident in all pressure signals, regardless of sensor type, as shown in Fig. 4.4 b). At these levels, acceleration sensitivity poses no significant problem that could not be adequately addressed through digital signal processing (i.

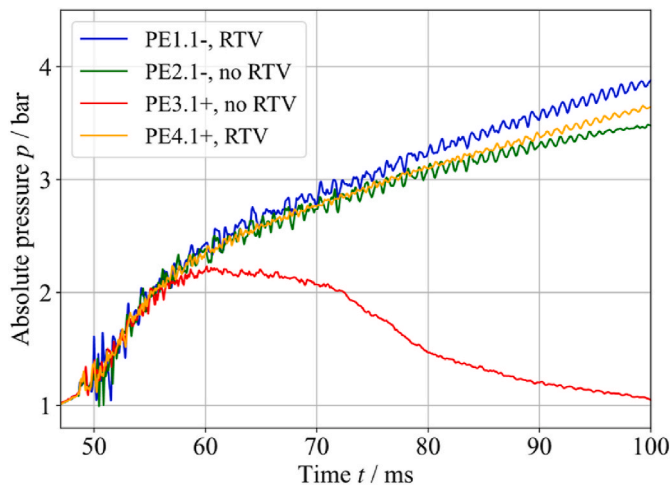


Fig. 4.3. Deviation of PE sensors (A-TR).

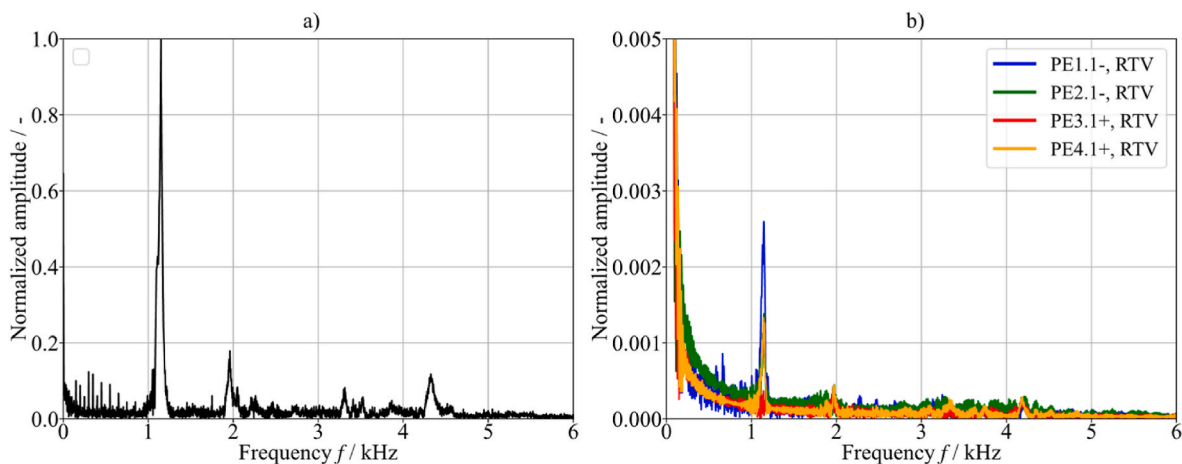


Fig. 4.4. a) Frequency spectrum of the PE5.1+ sensor (blind-hole) with H₂-gas; b) the frequency spectrum from PE1.1- to PE4.1+ with H₂-TR.

e., low-pass and/or band-pass filtering). However, extreme cases may occur where signal distortion prevents meaningful interpretation. The PE sensor 601CAA is particularly susceptible to this issue. As demonstrated in Fig. 4.5, compared to model 6031 (black), the higher acceleration sensitivity of the 601CAA (blue) results in greater signal amplitudes, i.e., more pronounced oscillations. When using a test enclosure such as the one presented here, pressure sensors with low acceleration sensitivity are preferable to minimize interference. PR sensors typically exhibit such low acceleration sensitivity, with both models 4080B and 4011A being suitable candidates for this application.

4.4. Acoustic resonance

In addition to enclosure vibrations, acoustic oscillations may occur. These originate not from the sensor's physical operating principle, but from cavities in front of the sensing diaphragm. As explained in Section 3.3.3, such cavities can induce acoustic oscillations that may cause resonant amplifications. While flush-mounted sensors typically avoid this phenomenon, the PR model 4080B inherently contains a design-related cavity (see Fig. 3.1, a)). Here, the diaphragm is recessed several millimeters behind the thread end, creating an empty volume. This cavity generated high-frequency interference signals in all

experiments, which superimposed on the actual measurement signal. Therefore, all subsequent discussions apply exclusively to this sensor model. For processes with longer PRT (P-TR, A-TR, and N-TR), only slight oscillations appear in the measurement signals leaving peak pressure detection unaffected.

Acoustic resonance becomes problematic only for rapid pressure rises (i.e., highly dynamic conditions). This includes all tests involving H₂-air explosions (with/without TR). Fig. 4.6 illustrates this: a) shows the time-resolved pressure profiles of stoichiometric H₂-air explosions with (H₂-TR, blue) and without simultaneous TR (H₂-gas, black), while b) displays the normalized frequency spectra from all four PR model 4080B sensors during H₂-TR. Pressure spikes can be observed in both cases (see Fig. 4.6 a)), red arrows). In H₂-air mixtures, peak amplitudes can exceed 1 bar, thereby significantly distorting the measured maximum pressure toward higher values. When these spikes occur near the pressure curve maximum, recorded values may surpass the actual peak pressure (see Fig. 4.6 a)): 8.3 vs. 7.9 bar). However, since these spikes do not appear in the strain gauge measurements during maximum deformation (Fig. 4.7, black) and no corresponding material stress is observed, they represent measurement errors rather than actual pressure.

An interesting aspect is the hypothesized origin of this interference. While the acoustic oscillations correspond to harmonic vibrations, the pressure spikes do not. Instead, they appear to result from a superposition of multiple oscillations, likely due to interference with the enclosure resonance vibrations discussed in Section 4.3. The enclosure resonance, being inherently low-frequency (see Fig. 4.4, a)/b)), develops gradually due to the substantial inertia of the flameproof enclosure. Because of the rapid pressure rise characteristic of H₂-air explosions, the enclosure only begins oscillating as the pressure approaches its maximum. Additionally, the amplitudes of the acoustic oscillations are significantly more pronounced due to the short PRT. An interference of slow enclosure resonance vibrations and high-frequency large-amplitude acoustic oscillations could thus cause amplification of the measurement signal. The fact that this is a superposition phenomenon involving both low- and high-frequency components complicates effective signal filtering. As Fig. 4.6 b) demonstrates, the acoustic oscillations do not exhibit distinct frequency peaks, instead producing a broadband signal whose frequency range shifts with the gas mixture's molar mass (see Section 3.3.3).

- stoichiometric H₂-air (avg. molar mass $\sim 20.6 \text{ g} \cdot \text{mol}^{-1}$): 4–5 kHz
- air, N₂, and stoichiometric C₃H₈-air ($\sim 29.0, 28.0, 29.7 \text{ g} \cdot \text{mol}^{-1}$): 3–3.6 kHz

Consequently, customized filters are needed for each experimental

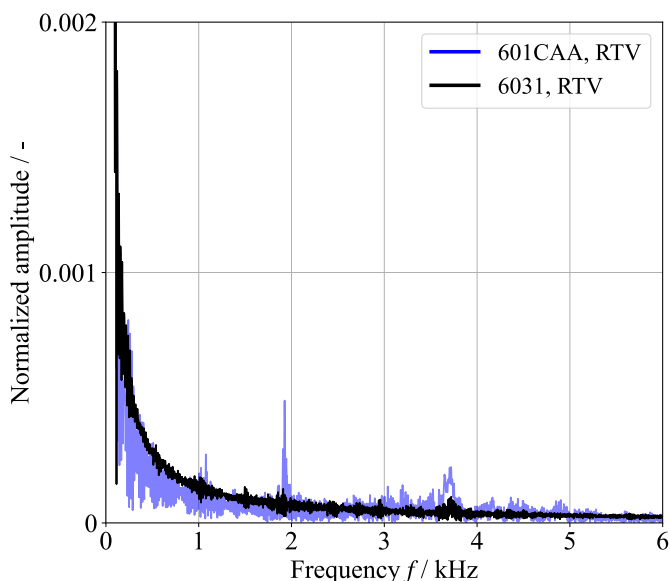


Fig. 4.5. Comparison of FFT spectra for 601CAA (blue) and 6031 (black) during H₂-TR.

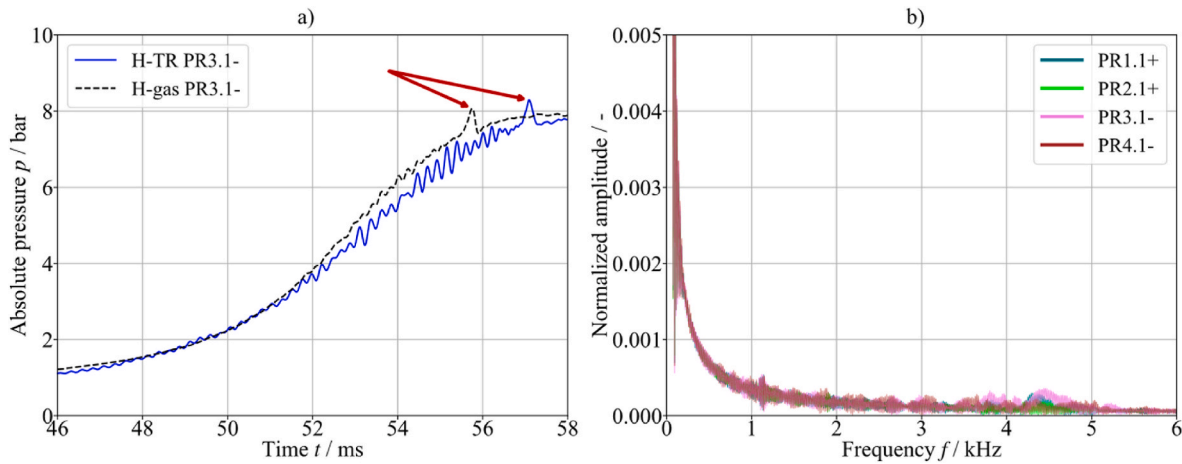


Fig. 4.6. a) Time-resolved pressure profile from PR sensor PR3.1- during H₂-TR (blue) and H₂-gas explosion (black); b) Normalized frequency spectrum of PR1.1+ through PR4.1- during H₂-TR. Pressure spikes are indicated by red arrows.

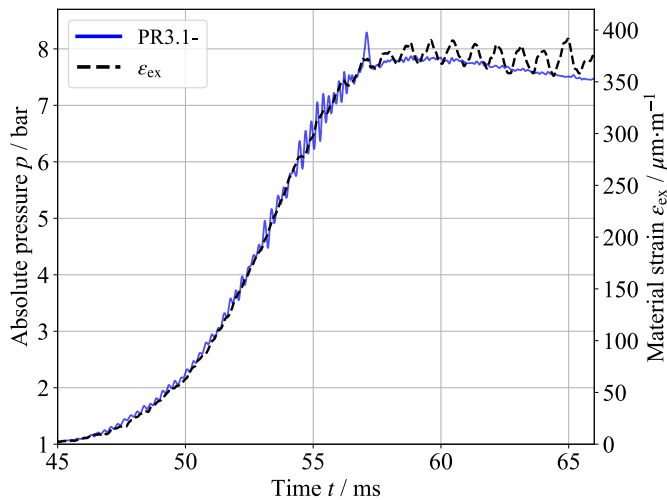


Fig. 4.7. Time-resolved pressure (PR3.1-, blue) and strain (black) during H₂-TR.

configuration to address both enclosure resonance and acoustic oscillations, making post-processing more complex. Visual inspection of pressure profiles is therefore mandatory for rapid rises when using non-flush sensors, as uncorrected peak pressures may otherwise be overestimated. An alternative is the use of flush-diaphragm sensors such as the model 4011A (PR5.2-), which eliminate acoustic resonance artifacts, as demonstrated in tests with and without TR.

4.5. Measurement robustness

Besides challenges inherent to the measurement task itself, the robustness of the sensor setup also varies significantly. While PR sensors generate measurement signals in the millivolt range, PE sensors merely produce signals on the order of picocoulombs. This makes the entire measurement chain - comprising the sensor, cabling, and charge amplifier - far more susceptible to all types of interference. Consequently, PE sensors and their associated equipment impose stricter technical requirements and are inherently more sensitive than PR sensors (see Section 3.3). This sensitivity is a disadvantage for harsh measurement environments, such as during LIB thermal runaway. Unless the measurement task specifically demands the high dynamic response of PE sensors, PR sensors are generally preferable due to their superior robustness.

5. Conclusions

This work systematically evaluates piezoelectric (PE) and piezoresistive (PR) pressure sensors to identify the most suitable technology for LIB TR pressure measurements. The main findings are:

- PR sensors exhibited consistent pressure curves and peak values with no positional dependence, which in all cases exceeded those recorded by the PE sensors. Consequently, the PR sensor signals were considered as reference pressure.
- PE sensors recorded lower values at the LIB's positive terminal than at the negative terminal side due to thermal shock.
- PE sensor signal reduction correlates with high thermal load of TR
- accurate peak pressure measurement with PE sensors was not possible, even with RTV and short PRT (H-TR)
- both sensor types exhibited oscillatory interference signals due to the sensors' acceleration sensitivity
- PE sensors, especially model 601CAA, are prone to severe distortion, rendering the measurement signal unusable.
- low acceleration sensitivity of PR sensors proves advantageous in setups with high mass and inertia

The decisive factor for the choice of pressure sensor is thermal shock sensitivity, rendering PE sensors generally unfit for pressure measurements during LIB thermal runaway. It is important to note that although the motivation for the presented experimental series originates from the field of explosion protection, the TR behavior of a cylindrical cell is consistently as described in this work. When measuring pressure during LIB TR, the choice should therefore generally fall on a PR pressure sensor with a flush-mounted diaphragm. Attention should be given to membrane flushness to avoid parasitic acoustic oscillations. It should also be noted that only oil-filled PR sensors were examined in this study. It is possible that other PR sensor technologies may behave differently from the results observed in this study. For an initial assessment, the flexible measurement range of PE sensors is advantageous, since a PR sensor could be destroyed if the pressure is underestimated. However, the experiments presented so far involve only simple enclosure geometries. With increasing complexity of the setup, more rapid pressure rises up to pressure piling may occur [27]. In such cases of high signal dynamics, PE sensors are traditionally used. In the future, the applicability of PR sensors, especially due to their robustness, would be of interest for these conditions. Furthermore, only cylindrical cells were examined, whose behavior during TR is not uniform (positive vs. negative terminal). Other cell geometries, such as pouch or prismatic cells, could produce different results.

CRedit authorship contribution statement

Freyja Galina Daragan: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohsen Shekoochian:** Methodology, Investigation, Data curation. **Stefan Essmann:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Tim Krause:** Writing – review & editing, Supervision, Resources, Conceptualization. **Ricardo Rodrigo:** Writing – review & editing, Supervision, Resources, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Perplexity AI, DeepSeek and DeepL Translator in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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